Estimation and Planning

FINTECH 512
Software Engineering

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Warmup Exercise

Estimate the cost and elapsed time required to make the following trip:

1. Travel, in person, from the starting point to the destination
   - Starting point: Traffic circle in front of Duke Chapel,
     - Address: 401 Chapel Dr, Durham, NC 27708
   - Destination: Smithsonian National Museum of Natural History
     - Address: 10th St. & Constitution Ave. NW, Washington, DC 20560
   - Start time: 6am tomorrow
2. Buy something from the gift shop.
3. Return to the starting point.

Budget: $2000. You keep the change.

Questions:

- How many days and hours would it take to complete the trip?
- How much would the trip cost?

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Trip Exercise review

- Questions:
  - How did you make the trip?
    - Drive, fly, bus, railroad, mixed modes?
  - How many days and hours would it take to complete the trip?
    - Shortest trip?
    - Longest trip?
  - How much would the trip cost?
    - Lowest cost?
    - Highest cost?
  - If your time was unlimited, how cheaply could you make the trip?
  - If your funds were unlimited, how quickly could you make the trip?

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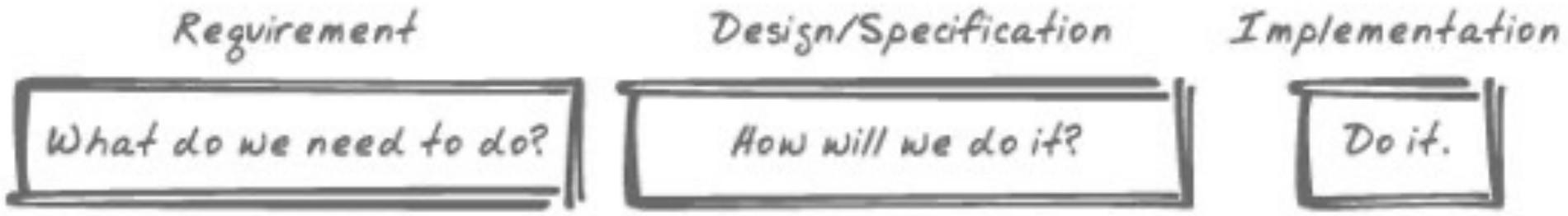
Project Planning



- Project **planning** involves answering several questions:
  - "What do we need to do?" (Requirements)
  - "How will we do it?" (Specification and Design)
  - "**How long will this take and how much will this cost?**"
- **Estimation** is the process of reasoning about how long it will take and how much it will cost.

[0] "Making Things Happen", Berkun

Estimation Quiz

- Without looking the values up, choose **upper** and **lower** bounds that give you a 90% chance of including the correct value for:

1. **Surface temperature of the Sun**
2. Latitude of Shanghai
3. Area of the Asian continent
4. The year of Alexander the Great's birth
5. Total value of U.S. currency in circulation in 2004
6. Total volume of the Great Lakes
7. Worldwide box office receipts for the movie Titanic
8. Total length of the coastline of the Pacific Ocean
9. Number of book titles published in the U.S. since 1776
10. Heaviest blue whale ever recorded

'Software Estimation: Demystifying the Black Art', Steve McConnell

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Estimation Quiz

- Without looking the values up, choose upper and lower bounds that give you a 90% chance of including the correct value for:

1. **Surface temperature of the Sun** (**10,000F, 6,000C**)
2. Latitude of Shanghai (**31N**)
3. Area of the Asian continent (**17,139,000 m, 44,390,000 km**)
4. The year of Alexander the Great's birth (**356 BC**)
5. Total value of U.S. currency in circulation in 2004 (**$719.9 Billion**)
6. Total volume of the Great Lakes (**5,500 cu miles, 23,000 cu kilometers**)
7. Worldwide box office receipts for the movie Titanic (**$1.835 Billion**)
8. Total length of the coastline of the Pacific Ocean (**84,300 miles, 135,663 km**)
9. Number of book titles published in the U.S. since 1776 (**22 million**)
10. Heaviest blue whale ever recorded (**380,00 pounds, 170,000 kilograms**)

'Software Estimation: Demystifying the Black Art', Steve McConnell

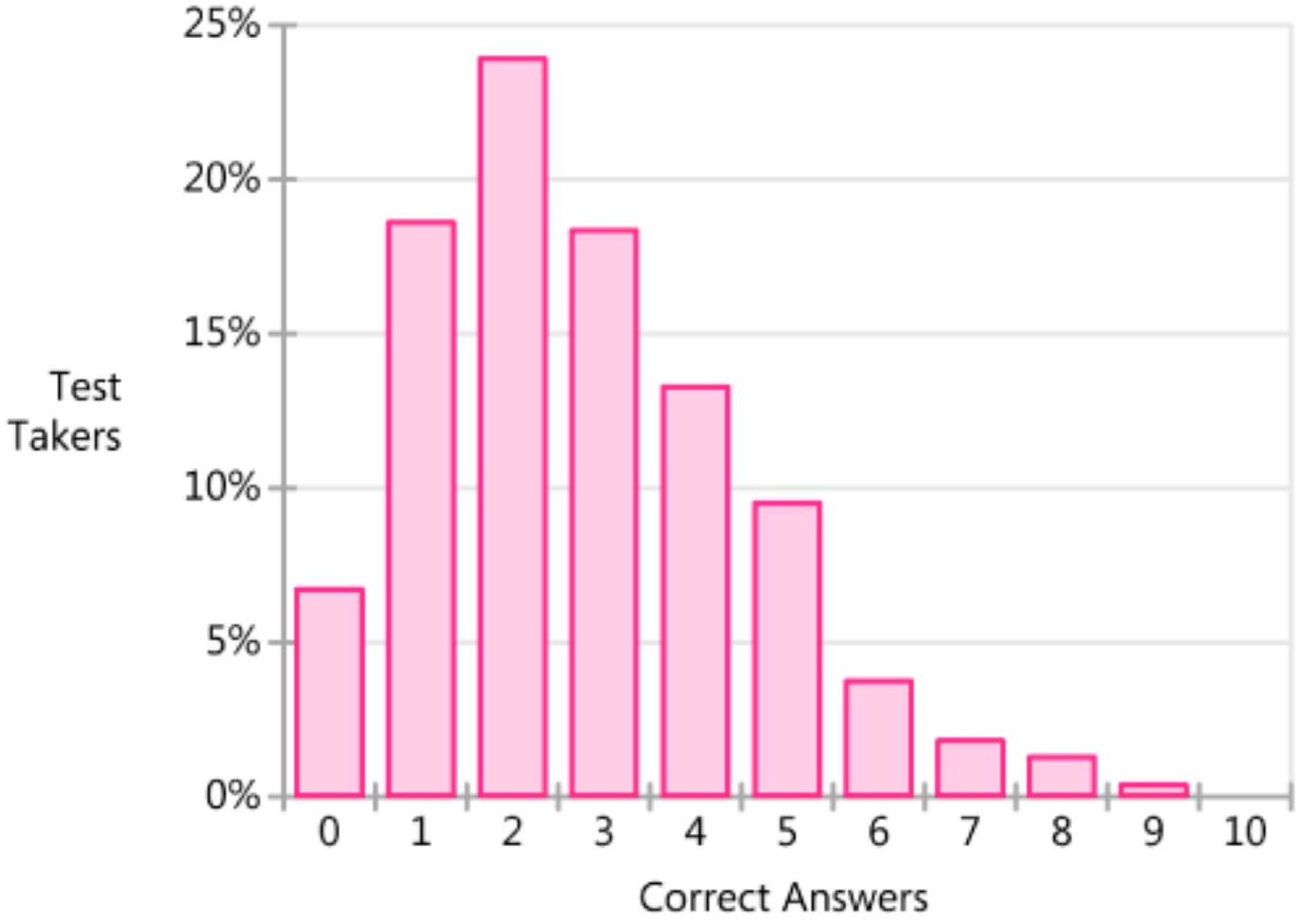

Figure 2-1. Results from administering the "How Good an Estimator Are You?" quiz. Most quiz-takers get 1–3 answers correct.

'Software Estimation: Demystifying the Black Art', Steve McConnell

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Quiz Assessment.. Estimation is hard

- Most people get 2-3 correct
  - They're more confident of their confidence intervals than they should be!

- Bias: Do you tend to overestimate or underestimate?
- Variability: Were some of your estimates close and some far off?

For the quiz questions, the answers can be looked up. For new software development, the answers often don't exist yet.

Conclusion: For good estimates, you need something more solid than guesswork.

'Software Estimation: Demystifying the Black Art', Steve McConnell

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What could go wrong?

- Inaccurate information about the project being estimated
  - Requirements
  - Technology
  - Environment
  - Organization
- Inaccurate information about the capabilities of the people that will perform the project
- Project 'chaos':
  - Unclear or unstable requirements
  - Poor design -> to many defects
  - Lack of experience
  - … and many more
- Inaccuracies arising from the estimation process itself
- Parkinson's Law: the work will expand to fill available time
- 'Student Syndrome' (Goldratt): "If developers are given too much time, they'll procrastinate until late in the project."

'Software Estimation: Demystifying the Black Art', Steve McConnell

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Solutions

- Bias
  - Recognize and account for your own biases
  - Use historical data to compute and apply a correction factor.
  - Compare your estimates to the actual values and with practice learn how improve your accuracy.
- Variability
  - Take advantage of “the law of large numbers”; more estimates, though each is imprecise, will be better, not worse! (aka, “Wisdom of Crowds”
  - Decompose the problem so you have to make more estimates.
- **Planning**
  - Estimates are more accurate when done based on a clear and accurate idea of how the requirements will be met.
  - But how do you get clear and accurate data before you've built the system?
  - Perform estimates throughout the process. Estimates should become more refined and closer to the actual value the later in process the estimate is made.
  - The better your answer for "How will we do it?", the better your estimates will be.

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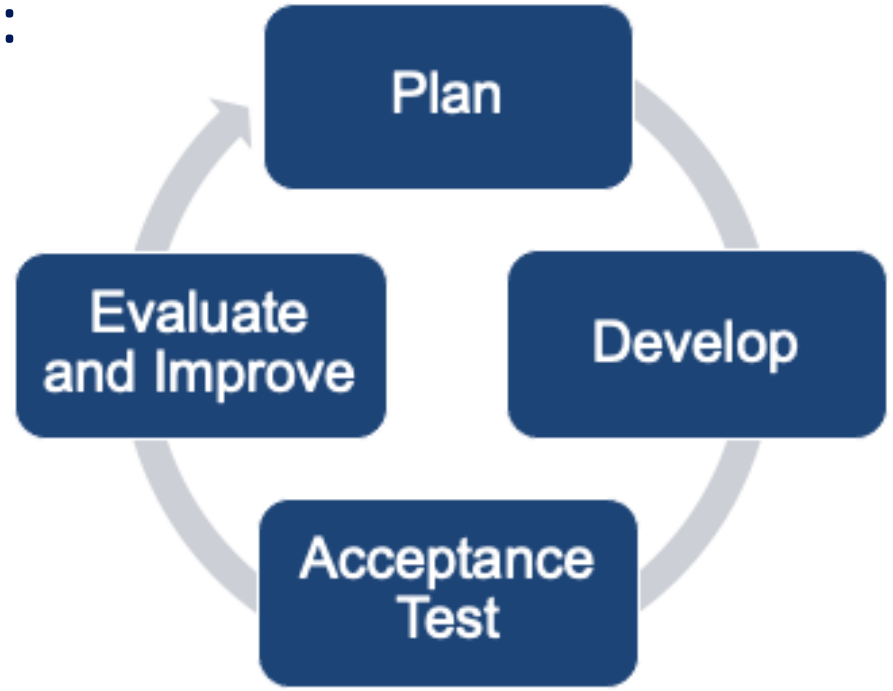
Personal Software Process (PSP)

PSP introduced in 1994, updated recently, breaks the discipline into the following skills:

- following a defined, measured process
- estimating work consistently
- reviewing work products effectively
- documenting detailed design



[0] '*A Discipline for Software Engineering*', Watts Humphreys

[1] 'PSP for the 21st Century', https://resources.sei.cmu.edu/library/asset-view.cfm?assetid=540135

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Personal Software Process (PSP)

Premises

- User perception of software value is governed by the quality of the software.
- The quality of a software system is governed by the quality of its worst components.
- The quality of a software component is determined by the knowledge, discipline, and commitment of the developer.

Outcomes

- Software...
  - Done better and done on-time
  - Fewer defects
  - Predictable delivery
- Developer practices...
  - do it right (do it well)
  - do it all (don’t skip steps)
  - do it once (minimize rework)

[0] '*A Discipline for Software Engineering*', Watts Humphreys

[1] 'PSP for the 21st Century', https://resources.sei.cmu.edu/library/asset-view.cfm?assetid=540135

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Problem: How do I complete a non-trivial task with minimal error?

Even small programs require:

- specifying the solution unambiguously
- blocking out time to work
- selecting programming languages, libraries, and reusable parts
- writing and testing code
- interacting with revision control

PSP Claim: With a personal process…

You will

- understand your ability
- be better able to improve
- learn a professional discipline

You can

- estimate and plan your work with increasing accuracy
- meet your commitments
- resist unreasonable commitment pressures

You will have the data to continually improve the productivity, quality, and predictability of your work.

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Plan in Enough Detail to be Useful

- If you do not understand what the product looks like, then you do not know enough to make an estimate.
- Good size estimates are detailed because that assures most needs are accounted for, and the parts are small enough to understand.
- Early estimators usually cannot think in detail because requirements do not uniquely map to a product.
- •The basic issues are
  - conceptually mapping all requirements to the product
  - our limited capacity to visualize the detail
  - keeping track of all the parts

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PSP approach: Estimation based on size

- Translate requirements into list of tasks
- Break large tasks down into tasks small enough to estimate
- Estimate each small task, add up aggregated tasks to find total
  - How long a programming task will take depends, at least in part, on how big the program will be: Effort = Size * Rate.

In your experience, do you think size is a good way to estimate time?

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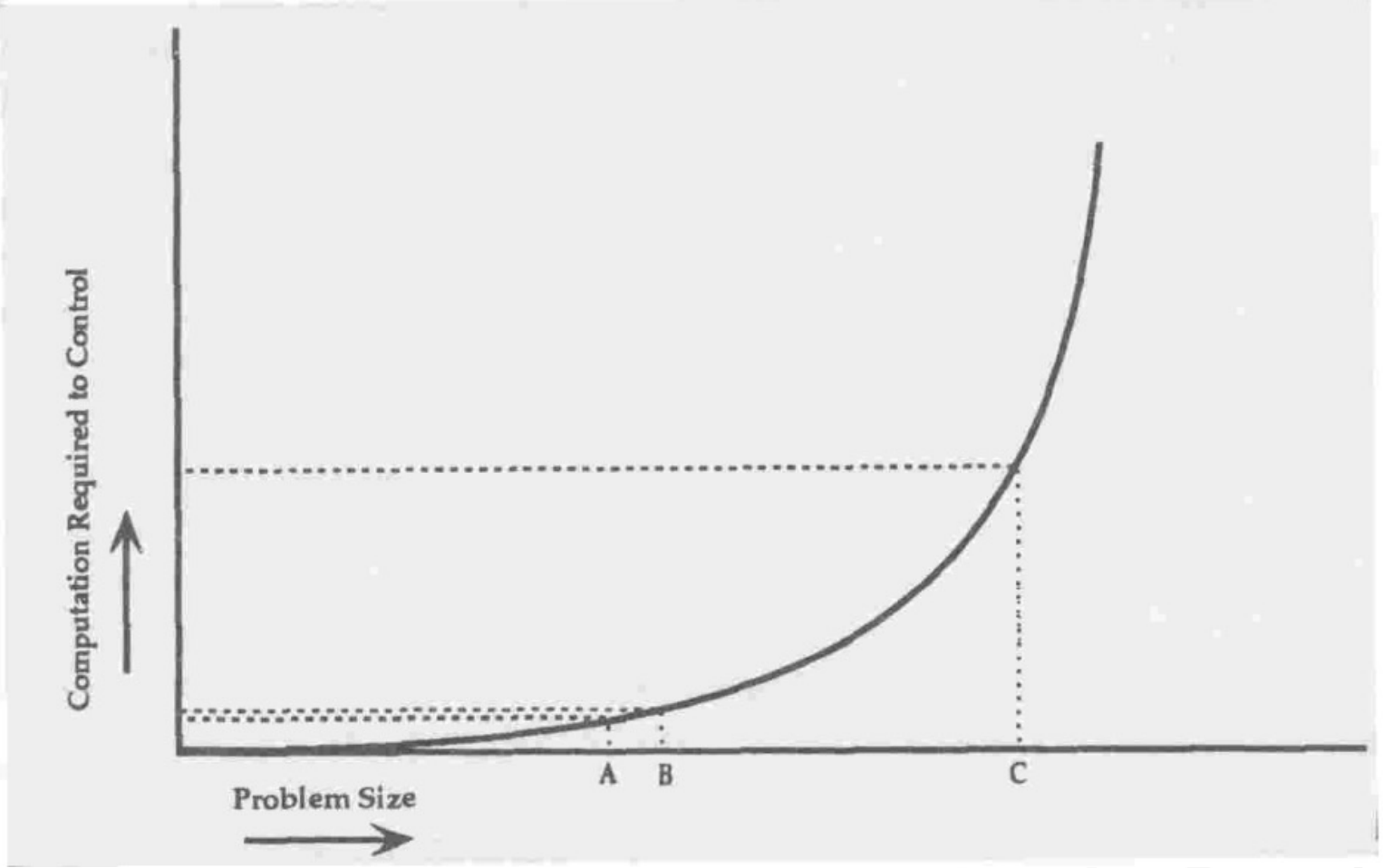
Size vs. Effort



'How Software Is Built', Gerald Weinberg

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Problem: Estimating size before development

- Because development effort is related to the product size, if we know the size we can estimate the effort and schedule.
- The problem is that we cannot directly measure size until the product is nearly complete.
- Initially we have only requirements. Unfortunately, most requirements:
  - are not directly related to development effort
  - do not account for implementation variations such as programming language or reuse
  - are not easily (or automatically) countable

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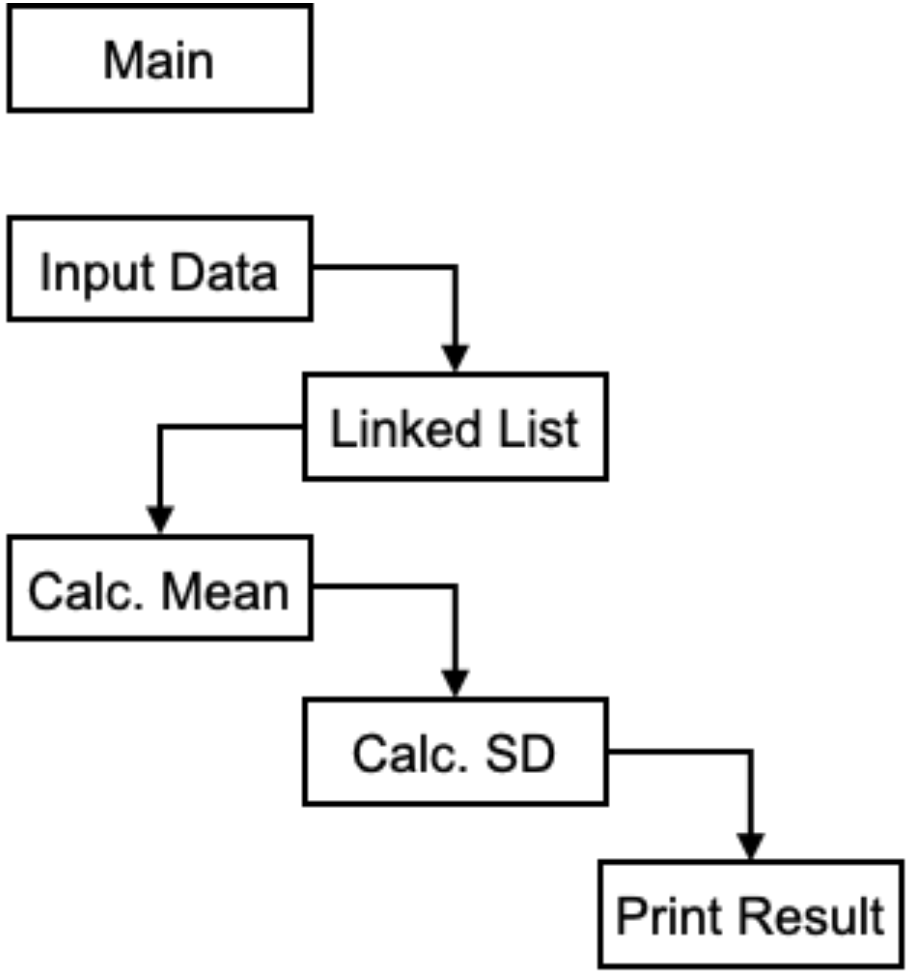
Solution: Decompose the Solution

- Decompose the problem to identify the implementable parts. User stories are one technique. We will use a **Planning Model** because we want to reason about the implementation.
- The Planning Model is notional. It identifies pieces that could be used to construct the program, if they existed.
- A Planning Model
  - conceptually relates the functional requirements to product parts
  - identifies the product elements that will produce the desired functionality
  - decomposes and refines the solution into sufficient detail to identify recognizable parts
  - records the parts so you do not forget anything

Example: Requirements

- Produce a program to take records of data from a file
- Store the data internally using a linked list
- Compute the mean and standard deviation
- Print the result to an output file

Example: Planning Model



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Example: Identify Base Parts to Modify or Reuse

Base Part	(B)ase size	(A)dded	(D)eleted	(M)odified
Main	50	10	5	5
Linked list	50	10	0	5

- Often, you begin with an existing program or parts.
- Some parts need only small changes, such as
  - deleting unused calls or processing
  - modifying variable names, loop conditions, or file names
  - adding new calculations
- For parts that you intend to modify,
  - measure the size
  - estimate the added, deleted, and modified lines
  - record your measures and estimates

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Example: Estimating the New Parts

Part	New Item	(B)ase Size	(A)dded	(D)eleted	(M)odified
Math package		150	10	0	0
Math package	Calculate SD	0	20	0	0
Math package	Calculate mean	0	10	0	0
Data input		0	50	0	0

- Other parts may be all new or may include new items such as added methods.
- Estimate the parts added to the base and new parts.

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Estimating Size from Planning Model

Estimate modifications to existing parts

Base Program	Size	Added	Deleted	Modified
Main	50	10	5	5

Estimate the parts added to the base and new parts

Base Additions (BA)	Type	Relative Size	Size
Data input	IO	M	20
Print output	IO	M	20

Assembly	Part	Type	Size	Size
Math package	Calc SD	Calculation	M	20
	Calc mean	Calculation	S	10
Linked list		Data	L	50

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Example: Completing the Estimate

- Estimate the **T**otal new program size **(T).** Start with the **B**ase lines of code, then add all the **A**dded lines of code, and finally subtract all the **D**eleted lines of code.
  - **T = B + A – D**
- Calculate the **E**stimated **new** code by summing the **A**dded and the **M**odified code from all base and new parts.
  - **E = A + M**

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Exercise ~~(Submit via Canvas, 'Estimation' link)~~

- Requirements: See use case on next slide
- Take 5-10 minutes to draft a rough implementation approach and time estimate:
  - Assume you are working with 'flask' based system.
  - Assume one person will do all of the implementation work
- Submit your estimate in Canvas
- Be prepared to present your solution in class
  - Describe the "pieces" required
  - Estimate how long each piece would it take

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Use Case Name:	Transfer Money to another student		
Use Case ID:	TR 1		
Created by:	John Smith	Last Updated By:	John Doe
Date Created:	7 January 2020	Last Revision Date:	10 January 2020

Description	Allows a card user to transfer money from their personal account to another individual
Actors	Student card holder
Preconditions	1. Student has authenticated into the BluePay system 2. Student has funds in their account to be able to transfer to another student.
Postconditions	1. Sending student's balance is lowered by the amount of the transfer. 2. Receiving student's balance is increased by the amount of the transfer. 3. Appropriate audit trail established for non-repudiation.
Flow:	From the BluePay main screen, the user selects the transfer money icon. The system will then show a list of "friends" that the user has selected as their favorites. The user then selects a friend to transfer money to. The user then enters the amount to transfer. The system validates the user has sufficient and transfers the funds once the user has confirmed.
Alternate Flows	If the user does not have any "favorites" or the desired recipient is not in that list, the user can search for a new recipient.
Exceptions	If the transaction logs are unable to be made, the system rolls back the entire transaction and notifies the user of the cancellation. If the user has insufficient funds, the transaction is cancelled and the user notified.
Assumptions	
Notes and Issues	

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After Development, Review and Calibrate

- For the total size, compare the total estimated to the actual size.
- For each component, compare the estimated relative size to the actual measured size.
- Where they differ, consider why your estimate was incorrect.
- The exercise of comparing your actual results to your estimate helps you calibrate your expert judgement. This calibration will improve your future estimation performance.

Software Units as Proxies for Size

- In the PSP, a proxy is a unit of software that can be identified early in a project.
- Proxies are normally based on functionality.
- Common proxies include screens, files, classes, objects, logical entities, functions, procedures, methods, and function points.
- Proxies can be identified formally or informally:
- Formal approaches include function point estimation.
- Informal approaches include creating user stories.
- PSP uses the Planning Model.

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Estimation is a learned skill

- Even simple plans are subject to error:
  - unforeseen events
  - unexpected complications
  - better design ideas
  - random variation
  - just plain mistakes
- The best strategy is to use a consistent method to plan in detail:
  - Identify the recognized tasks.
  - Make estimates based on similar experiences.
  - Make judgments on the rest.

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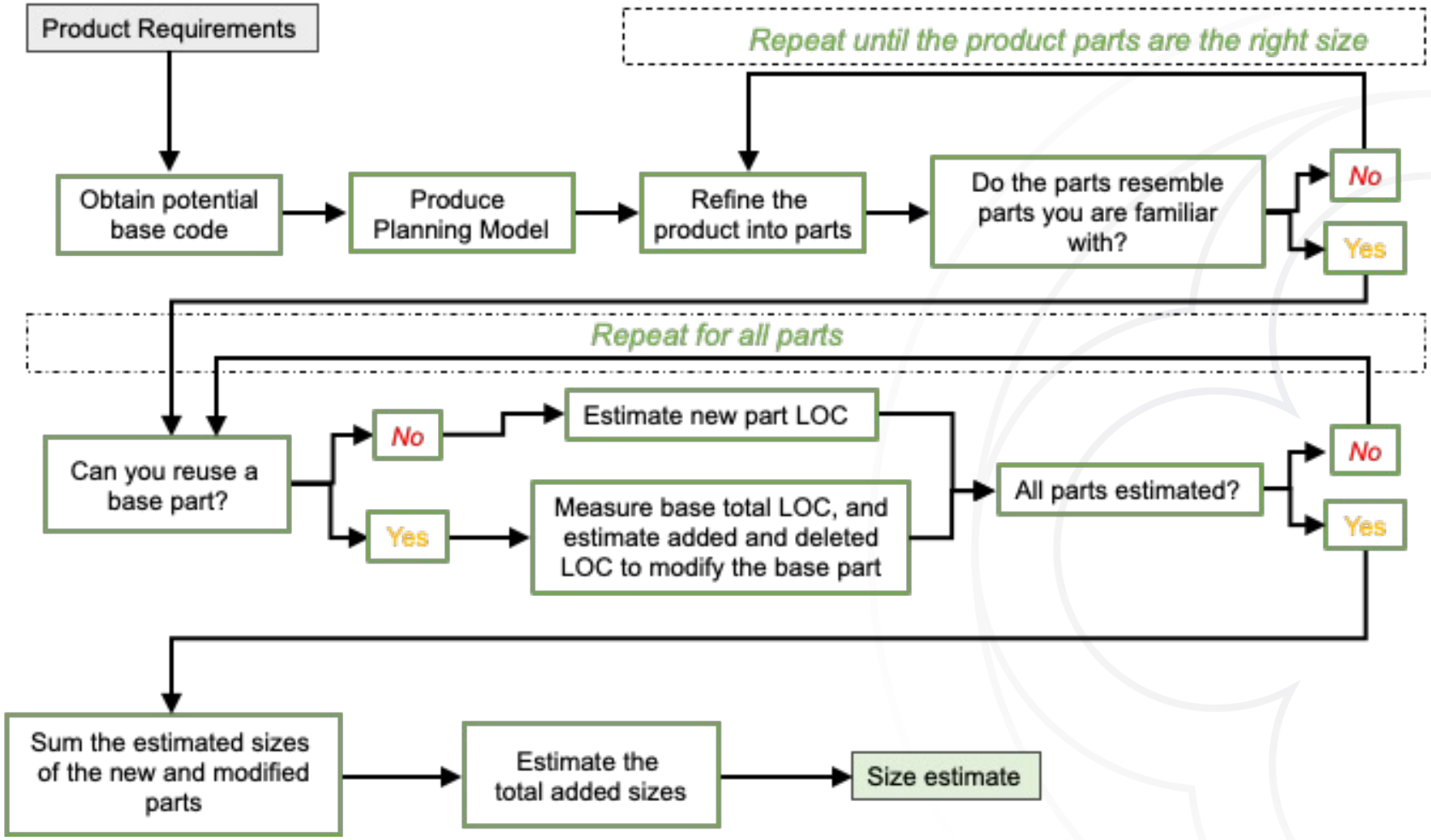
Size Estimating Overview Without Historical Data



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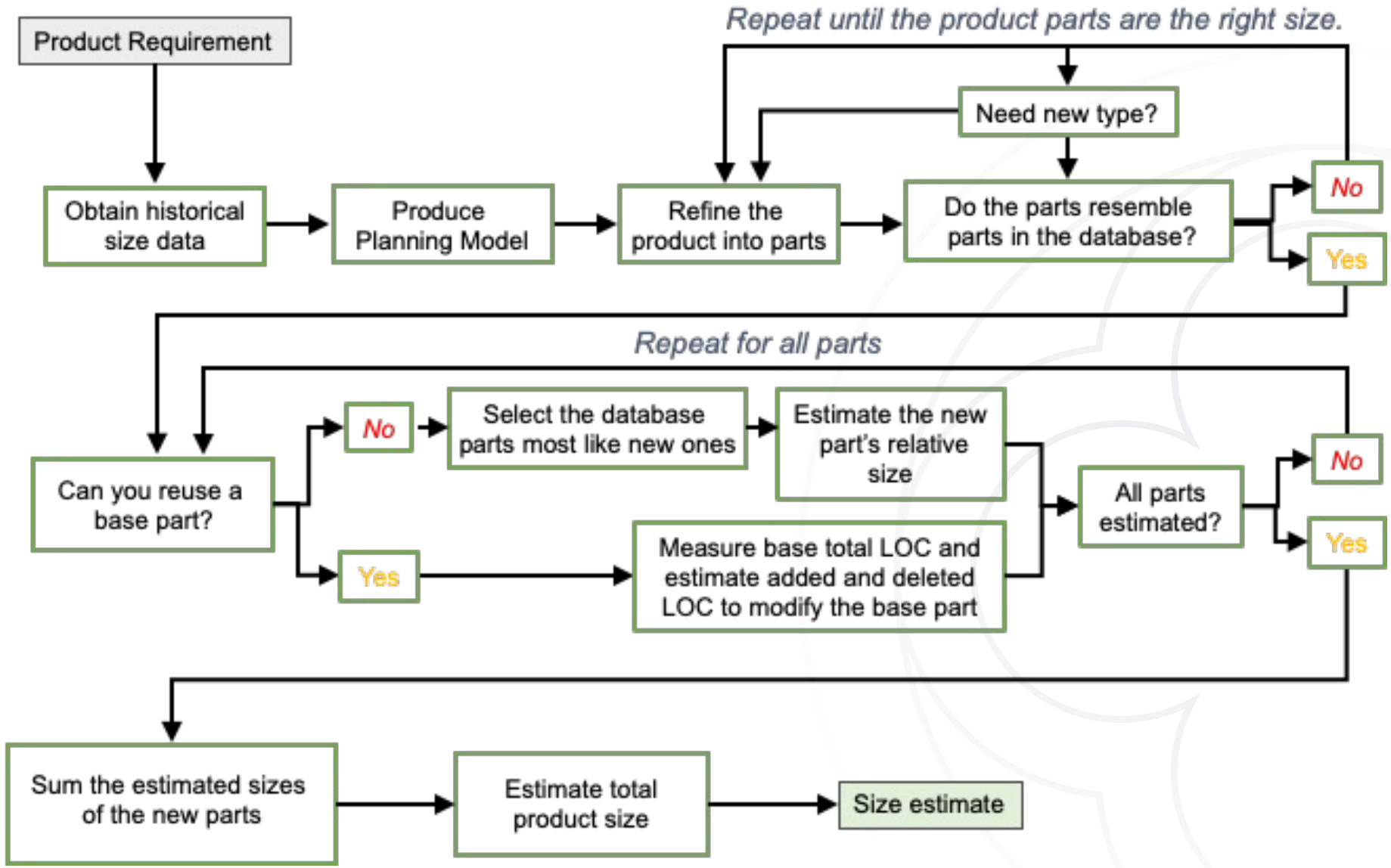
Size Estimating Overview



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